Microdosing of Strength Training in Basketball During Congested Game Weeks: A Practical-Methodological Approach

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Basketball is a team sport with intermittent characteristics whose motor actions performed by players require complex improvement of technical, tactical, physical and mental aspects to be successful.[1, 2, 3] According to Vallés-Ortega et al [4] basketball has a high physical demand. Therefore, the different specific biomotor capabilities must be stimulated throughout the season in order to maintain competitive readiness, improve athletic performance and reduce the incidence of injuries.[5] Systematic monitoring of training and competition loads is necessary to understand the training process and how to obtain the desired adaptive responses. The prescribed external load (volume, intensity, density, frequency and complexity of tasks) is individually controlled, and is adjusted according to the psychobiological behavior of the internal load (objective and subjective).[2, 4, 5] At this point, it is necessary to distinguish three types of physiological effects provided by the distribution of loads: acute (immediately after the training session and/or competition), delayed (after a few hours or days) and accumulated (after weeks and/or months). The correct mastery of these effects in the face of loads is what will allow fatigue to not set in and the physical condition to be improved.[4, 5] The planning in the distribution of loads is done by the pre-selected periodization model.

The dynamics of loads follows a sustainable content with a logical and rational tendency, seeking a progressive increase in physical fitness and recovery from psychobiological stressors.[4, 5, 6] It is not an easy task to deal with the stress of the loads imposed on the athletes' bodies. The competitive basketball scenario encompasses chaotic stressful situations such as constant travel, little time available for training, stages of incomplete recovery, high pressure from the media and managers, high expectations of results, constant changes of coaches and

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players, high incidence of injuries, among other complex factors to manage.[3, 7, 8] One of the real difficulties faced by strength and conditioning coaches is congested game microcycles. In these weeks, teams have to live with a high density of games according to their league calendar.[1, 3, 4, 9] In this perspective, reports point to a density between two to five games in a period of seven days in the NBA (National Basketball Association).[3] In the Spanish women's basketball league, in some situations, players face intervals shorter than 72 hours between matches.[4]

On the other hand, in the men's Euroleague, the tendency is to play between three to four games a week.[9, 10] In these critical periods, it is necessary to give multidimensional attention to an effective recovery from psychobiological fatigue so that the team avoids injuries and illnesses, as well as the players are in a ready condition to carry out a next training session or match.[1, 3, 8] Typically, in these periods of high volume of weekly games, teams tend to reduce training loads to compensate for the heightened stress.[1]

A valuable pedagogical resource in these situations is the so-called microdosing of training. In microdosing of training, the aim is to apply a minimal effective dose of physiological stimulus in order to maintain or improve physical condition.[5, 9] This study focuses on discussing the microdosing of strength training in the weeks of game congestion, since strength training and its functional manifestations in basketball players have a moderate to high correlation with the improvement of other biomotor capabilities such as speed and agility.[12, 13, 14, 15, 16, 17] In this regard, it is necessary to understand that stabilizing or improving the neuromuscular patterns of basketball players can be a determining element in athletic performance in situations with little time for training.

CHARACTERISTICS OF THE MICRODOSING OF STRENGTH TRAINING

An etymological analysis of the word microdosing shows that the prefix "micro" means small and the suffix "dosing" refers to dosage. When the prefix is added to the suffix, we find the term "small dosage".[18] Microdosing or microload of training has the function of maintaining or improving physical condition through a minimum effective dose of physiological stimulus.[5, 9] The microdosing of training fits as a crucial strategy in congested game weeks because it minimizes adverse side effects of high psychobiological stress, such as fatigue and injuries.[5, 19] Microdosing sessions have a short duration.[19] There is no consensus on what the "optimal duration" of a microdosing session would be, but it can be speculated that a total time period ranging from 15 minutes to 40 minutes (average values of approximately 20 minutes) would meet operational needs.[5] The key element in this complex equation is the

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gradual cumulative effect of small loading doses.[11] The loads used in microdosing sessions are low volume and high intensity. It seems that this combination of loads is better suited to congested microcycles to provide the minimum stimuli needed.[11, 20] The weekly frequency of microsessions of strength to stimulate physiological adaptations would be around 2 to 4 times a week.[6, 21, 22] Among the manifestations of strength that can be stimulated in microdosing of strength training sessions aimed at basketball players would be hypertrophy, maximum strength, power and power endurance.[5. 6, 9, 10, 23] The exercises that are prescribed are intended to increase or maintain body muscle mass (hypertrophy), maximal strength, power and power endurance.[5, 9, 10, 23, 24] It is also possible to use exercises that generate neuromuscular stimulation for activation of muscle groups as a preparatory exercise activity.[25] Another alternative is the use of complex exercises that provoke both acute and chronic neurophysiological effects, such as post-activation potentiation and post-activation performance enhancement, respectively.[24] Some experts advocate using a total of five to seven exercises in microsessions. However, this amount of exercises can vary substantially depending on the objective.[6, 9]

Regardless of the number of exercises, what should be considered is the execution of multiarticular and multiplanar primary exercises.[10, 25] From this perspective, the number of sets and repetitions are linked to the manifestation of the strength that will be stimulated.[5, 22] The timing of application of microdosing of strength training should be analyzed with caution, as there is no research to prove the most appropriate time to use microdosing in the congestion microcycle. The literature reports certain specific moments when microdosing of strength training can be used: 1) a few hours before the warm-up, 2) in the warm-up, 3) immediately after the match or 4) any period between games.[5, 9]

The use of microdosing hours before the warm-up occurs with the neuromuscular optimization function based on the ergogenic effect of the circadian rhythm. For example, athletes are stimulated with a microsession of strength training in the morning, aiming to obtain performance gains in the afternoon when the game takes place.[26] In the warm-up itself, microdosing acts as a stimulating component of some of the force's manifestations.[5, 27] After the match, microdosing can be used as compensatory training for players who have played fewer minutes on the court and can also be used to regenerate the body as an active recovery to remove metabolic waste.[6, 28] In other free time periods, during game congestion weeks, microdosing of strength training is seen as a quick training session that seeks to maintain or improve some component of functional strength. If the microdosing of strength training is well designed, a sensible balance between load and recovery can be maintained in the high-density game schedules that are present in competitive basketball.[1, 5]

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Tables

Table 01. Basic characteristics of the microdosing of strength training sessions for basketball players

VARIABLES	CHARACTERISTICS
Duration of the microdosing session	~15-40 minutes (average: 20 minutes)
Loads	Low volume and high intensity
Minimum weekly frequency	2-4 microsessions
Manifestations of strength that can be stimulated	Hypertrophy, maximal strength, power and\or power endurance
Moments of microdosing application	Before matches, during warm-up, after matches, and \or periods between games

Table 02. Types of exercises that can be employed in microdosing of strength training sessions

TYPES OF EXERCISES	OBJECTIVE
Muscle activation	Neuromuscular activation as a preparatory exercise
Complexes	Post-activation potentiation (PAP) and Post-Activation Potentiation Enhancement (PAPE)
Hypertrophy	Improve or maintain body size
Maximal strength	Improve or maintain maximal strength levels
Power	Improve or maintain power levels
Power endurance	Improve or maintain power endurance levels

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